Updated growth analysis for Alaska sablefish

Dana Hanselman[1] and Katy Howard[2]

[1] *NOAA Fisheries*

[1] *University of Alaska-Fairbanks, Fisheries Division*

1.0 EXECUTIVE SUMMARY

Growth parameters for Alaskan sablefish have not been updated for stock assessment purposes since Sasaki (1985). Meanwhile, many more sablefish have been aged with better geographic coverage. In this study we updated and corrected for bias in the older length-stratified data (1981-1993), analyzed newer randomly collected samples (1996-2004), and estimated new length-at-age and weight-at-age parameters. We then applied the updated growth data to the current stock assessment model. Our analyses showed that both male and female sablefish growth has changed significantly. Recently, sablefish are growing to a moderately larger maximum size. For use in the 2008 sablefish stock assessment, we recommend using the updated growth information divided into the two time periods (1981-1993 and 1996-2004). This new information provides the best fit to the data when applied in the stock assessment model and also provides results that are biologically reasonable.

1.1 INTRODUCTION

Growth parameters for Alaskan sablefish have not been updated since Sasaki (1985). When age-length transition matrices were first added to the stock assessment in 1995, they were constructed from data (1987-1993) that were collected under a length-stratified sampling scheme. These data were randomized by using the method of Kimura and Chikuni (1987), but these data were from limited areas and years and were aggregated in a way that put too much weight on large fish. Meanwhile, many more sablefish have been aged with better geographic coverage. Since the last update of sablefish growth, significant changes in length-at-age have been discovered for other species and have caused substantial changes in stock assessment results, such as with Pacific halibut (Clark et al. 1999).To evaluate whether changes to sablefish growth have occurred, we examined all the length-at-age data that has been collected on the longline survey since 1981. We then examined the sensitivity of the current stock assessment model to utilizing this new growth information by showing the effects of different growth scenarios on biomass trajectories.

1.2 METHODS

1.2.1 Length-at-age analysis

Length, weight, and maturity of sablefish specimens have been collected from the inception of the Japanese longline survey in 1978 and continues in the current NMFS domestic longline survey that started in 1987. These data were collected under two different sampling designs.

From 1981-1993, ages were sampled under a length-stratified design (a pre-determined number of otolith pairs were collected for each length). Estimates produced from length-stratified data create biased estimates of mean length at age for the population. This bias is caused by ageing smaller and larger specimens more often than would be aged under a random sampling design. This results in the mean size-at-age for early age groups to be too small, while the mean-size-at-age for the oldest age-groups is too large (Goodyear 1995, Sigler et al. 1997, Bettoli and Miranda 2001).

Fish aged 31 years and older were pooled together into a 31+ age category (Hanselman et al. 2006). In order to correct this bias in the length-stratified data (1981-1993), we considered the length data for all years to be a random sample from the longline survey and used the samples to create bias corrected age-length samples of the 1981 – 1993 data, using the following method (Bettoli and Miranda 2001):

$$\overline{L}_i = \frac{\sum N_j (n_{ij} / n_j) l_i}{N_i}$$

where $\overline{L}_i$ is the mean length-at-age, l_i is the length-at-age in subsample j, N_j is the number of fish in the j^{th} length-group, n_j is the number of fish subsampled in the j^{th} length-group, n_{ij} is the number of fish in the i^{th} age group and the j^{th} length group, and N_i is the number of fish in the i^{th} age group over all j length-groups. The von Bertalanffy (LVB) age-length model was fitted to bias corrected mean length at age data from 1981 – 1993 and to randomly collected age-length data from 1996 – 2004 by nonlinear least squares (Figures 3 and 4),

$$\overline{L}_a = L_\infty (1 - e^{-K(a - t_0)}) + \varepsilon_a$$

where ε_a is an additive error term, and L_∞, K, and t_o are model parameters. L_∞ represents the average maximum length, K describes the mean growth rate, and t_0 describes the mean theoretical age a fish would have been zero length (McDevitt 1990, Quinn & Deriso 1999). LVB growth curves were further fit to data by area and sex to look for differences in growth within the two time periods among different areas. The results by area are presented, but not discussed in this document; because the stock assessment is not subdivided into small areas.

Standard errors, correlation estimates, and 95% confidence intervals for each LVB growth curve parameter were estimated using the Hessian method. Individual parameters of growth models were compared between different data sets using the univariate Fisher-Behrens test, in which variance is not assumed to be constant (Quinn & Deriso 1999).

Hotelling T^2 multiparameter tests, analogues to the one-parameter, two-sample Fisher-Behrens test described above, were carried out to compare growth curves from different data sets. The Cerrato approach (Quinn & Deriso 1999) was used to avoid the assumption of common variance-covariance matrices (Quinn & Deriso 1999), as the difference of sample size between most data sets can be large. The variance-covariance matrix of $\Delta\hat{\theta}$ is now $\hat{\mathbf{V}}_3 = \hat{\mathbf{S}}_1 + \hat{\mathbf{S}}_2$ and the Hotelling T^2 test statistic is

$$T_3^2 = (\Delta\theta)' \hat{\mathbf{V}}_3^{-1} (\Delta\hat{\theta}).$$

The residual degrees of freedom for each data set is $f = n - p$ for a growth model with p parameters. The effective overall degrees of freedom, f, is

$$f^{-1} = \frac{1}{f_1}\left(\frac{(\Delta\hat{\theta})'\hat{V}_3^{-1}\hat{S}_1\hat{V}_3(\Delta\hat{\theta})}{T_3^2}\right)^2 + \frac{1}{f_2}\left(\frac{(\Delta\hat{\theta})'\hat{V}_3^{-1}\hat{S}_2\hat{V}_3(\Delta\hat{\theta})}{T_3^2}\right)^2 .$$

The test statistic T_3^2 is then compared against the critical value

$$T_{1-\alpha}^{*2} = \frac{fp}{f-p+1}F(p, f-p+1)_{1-\alpha} ,$$

where $f = n - 2p$ and $F(p, f\text{-}p\text{+}1)_{1-\alpha}$ at $\alpha = 0.05$ is the appropriate tabled F critical value. The null hypothesis is rejected if $T_3^2 > T_{1-\alpha}^{*2}$.

1.2.2 Weight-at-age

Weight-at-age data from the domestic longline survey was available from 1996-2004. This data was collected randomly and could be used directly without bias correction. To determine weight-at-age for the stock assessment model, first the length-weight relationship was determined using the typical nonlinear allometric relationship:

$$\hat{W} = \alpha L^{\beta} + \varepsilon$$

A common method to fit weight-at-age data is with the four-parameter LVB model. However, due to high parameter correlation with only one dependent variable, it is usually difficult to fit all four parameters at once, so a convenient method is to fix the allometric parameter β, determined from the length-weight relationship as a fixed parameter (Quinn and Deriso 1999). For this data set, there was a multiplicative error structure (Figure 1), so the log-transform of LVB model gives:

$$\ln\hat{W}_a = \ln W_\infty + \beta\ln(1 - e^{-K(a-t_0)}) + \varepsilon_a .$$

Nonlinear least squares estimation was used to determine W_∞, K, and t_0, with β fixed. These estimates of weight-at-age were then applied to the stock assessment model.

To obtain weight-at-age estimates for the older growth regime, we applied the newly estimated length-weight relationship to the bias-corrected length-at-age relationship. This was preferable to the observed average weight-at-age previously used. Also, the length-weight relationship would be expected to change much less than the length-at-age relationship.

1.2.3 Application to stock assessment

Length-at-age information is used in the Alaskan sablefish stock assessment through age-length transition matrices. These matrices are used to take estimated numbers-at-age in the model and predict lengths to compare with observed length compositions. If these matrices are developed with growth data that does not correspond with the true growth regime, then this can bias the model.

Data previously used in the model to populate the age-length transition matrices were raw, but randomized lengths-at-ages from 1981-1993 (Figure 2). A smooth growth curve based on more years and geographic coverage for the historical growth and a new growth curve for recent length-at-age should better describe the underlying population dynamics.

The age-length transition matrix is a matrix describing the probability of a fish of a given age to be a certain length. To develop this matrix, we use the estimated growth curve as the highest probability range of the matrix, but gave it normal error to account for the variability in length-at-age. The amount of error added to the growth curve is determined by fitting a nonlinear model to the observed standard deviations of each corresponding growth data set. This model takes the form:

$$\hat{s}_a = \alpha \ln(a) + \beta$$

where $\hat{s}_a$ is the estimated standard deviation of length-at-age at age a, α is a scalar parameter, a is the age of fish, and β is the intercept. Each age-group is weighted by its sample size in the nonlinear least squares procedure.

The resulting length-age transition matrices were then applied to the current stock assessment model. Preliminary results are briefly compared for four new model runs with the accepted assessment model from last year. The base model from 2006 (**0**), updated weight-at-age data (**1**), updated 1981-1993 data (**2**), only 1996-2004 data (**3**), or updated growth information for both time periods (**4**).

We attempted to separate the weight-at-age relationships into the same two time periods as above. In all cases, the model fit was worse than simply using the new weight-at-age data. Therefore, we only show model results using the 1996-2004 weight-at-age data.

1.3 RESULTS

1.3.1 Length at Age Analysis:

Results from the comparison of LVB growth curves fit to updated 1981 – 1993 data against 1996 – 2004 data for all Alaskan waters show similar results for both male and female sablefish (Figure 3, Tables 1 and 2): older (1981 – 1993) data fish display smaller asymptotic lengths (L_∞), slower growth rates (κ), and smaller t_o estimates. Results of the univariate Fisher-Behrens test on the male data show that only the L_∞ parameter is significantly different ($p < 0.01$) between the old (1981 – 1993) and new (1996 – 2004) data, but according to the multiparameter Hotelling T^2 statistical test, the two growth curves are significantly different ($p < 0.01$). Test results on the female data show that the L_∞ ($p = 0.00$) and t_o ($p < 0.01$) parameter estimates are significantly different, and that the two growth curves ($p < 0.01$) are significantly different as well.

Current average maximum length estimates used in the 2007 Alaska Sablefish Stock Assessment are 69 cm for males and 83 cm for females (Hanselman et al. 2006). Improving the estimates of the 1981-1993 data resulted in lower maximum size at age than that currently used in the stock assessment model. However, the newer data shows larger lengths-at-age then the updated older data. Parameter estimates by region and time period are shown in Tables 3 and 4.

1.3.2 Weight at Age Analysis:

Results of the length-weight and weight-at-age analysis are shown in Table 5. Average maximum weight was 3.16 kg for males and 5.47 kg for females. These estimates of maximum weight are smaller than the observed average values currently used in the stock assessment.

1.3.3 Stock assessment application

The model fitted the standard deviation of length-at-age data well (Figure 4). Using these relationships, we constructed new age-length transition matrices for the two time periods using the new growth curves described above. The resulting matrices (Figure 5) are smooth and more realistic than the rough matrices used in Hanselman et al. (2006, Figure 2).

When the updated weight-at-age data was applied to the model (**1**), the effect was a slight downward shift of the entire spawning biomass curve (Figure 6). This result is expected as the overall weight-at-age curve is slightly below the historic observed weight-at-age data previously in the model. When we exchanged the older observed length-at-age data with the bias-corrected LVB fit (**2**) the curve generally shifted up slightly from the base model (**0**) mainly at the peak biomasses, and substantially from Model **1**. When we apply only the new growth curve (1996-2004) to the model (**3**), this results in a dramatic downward revision of estimated spawning biomass (Figure 6).

Finally, when we choose two growth regimes and apply both growth data sets (**4**), the estimated spawning biomass series returned to similar magnitudes as the base model (**0**) (Figure 6). The principal difference between model **4** and the base model (**0**) was that the lows in estimated spawning biomass were lower, and the initial biomass in 1960 was quite a bit higher. This result of model **4** is logical because it would be expected that initial biomass should be high because fishing mortality was low prior to 1960. Another pertinent difference between model **4** and the base model is that although the estimated spawning biomass in recent years is slightly lower in model **4**, the upward slope of the recent trajectory is steeper than in the base model. Therefore, Model **4** will likely yield similar harvest recommendations to the base model in the near future.

Updating the growth data in general improved the fit to the data (Figure 7). Changing the weight-at-age data improved the fit to the data slightly, but adding the bias-corrected age-length matrices only from the older data (**2**) yielded a fit similar to the base model (**0**). Using the newer growth data (**3** and **4**), yielded substantially better fits than the base model (**0**), and splitting the growth into two time periods (**4**) yielded the best fit to the data.

1.4 DISCUSSION

Our analyses show that there has been some change in the growth of both male and female sablefish. While these changes were not severe, they were significantly different. It appears that recently sablefish are growing to a larger maximum size.

For use in the 2008 sablefish stock assessment, we recommend using the updated growth information divided into the two time periods. Not only does it provide the best fit to the data, it provides results that are biologically reasonable. The choice of where to split growth regimes was not based on a visible shift in growth at that time, but on a change in sampling design on the longline survey. Separating these data periods buffers the model from any other unforeseen effects that the sampling design change may have had on the data besides the bias expected on the tails of the distribution. The addition of the newest data will be more biologically realistic, while only having nominal effects on harvest rates.

1.5 REFERENCES

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Table 1. Male sablefish length-at-Age LVB parameter estimates and test results.

	1981 - 1993		1996 - 2004			Univariate Tests			
	MLE	SE	MLE	SE	$\Delta\theta$	z	f_e	P-value	Conclusion
L_∞	65.269	0.341	67.774	0.127	-2.50	6.882	35	0.000	Reject null
k	0.227	0.029	0.290	0.009	-0.06	2.033	32	0.051	Fail to reject
t_o	-4.092	0.936	-2.273	0.171	-1.81	1.911	29	0.066	Fail to reject
RSS	34		91,089						
n	30		4,889						

Multivariate Test

T_3^2	183.071
P-value	0.000
f_e	36.41
num df	3
den df	34
Fcrit	2.883
T^{*2}	9.150
Conclusion:	Reject null

Table 2. Female sablefish length-at-Age LVB parameter estimates and test results.

	1981 - 1993		1996 - 2004			Univariate Tests			
	MLE	SE	MLE	SE	$\Delta\theta$	z	f_e	P-value	Conclusion
L_∞	75.568	0.460	80.220	0.221	-4.65	9.116	42	0.000	Reject null
k	0.208	0.018	0.222	0.005	-0.01	0.758	33	0.454	Fail to reject
t_o	-3.629	0.523	-1.949	0.119	-1.67	3.131	31	0.004	Reject null
RSS	73		191,866						
n	31		5767						

Multivariate Test

T_3^2	164.595
P-value	0.000
f_e	40.18
num df	3
den df	38
Fcrit	2.852
T^{*2}	9.003
Conclusion:	Reject null

Table 3. Estimated LVB growth curve parameters for male sablefish stratified by region and time period in Alaskan waters.

Area	Time Frame	L_∞	κ	t_0	RSS	n
Chirikof Slope	1981 – 1993	70.863	0.226	-2.587	318.449	26
	1996 - 2004	67.272	0.335	-1.617	4430.503	294
Aleutian Slope	1981 - 1993	67.536	0.170	-6.255	105.491	30
	1996 - 2004	67.723	0.237	-3.323	4650.722	285
Kodiak Slope	1981 - 1993	69.615	0.138	-7.413	529.277	29
	1996 - 2004	66.595	0.357	-2.052	9195.078	542
Shumagin Slope	1981 - 1993	65.699	0.328	-1.658	207.361	30
	1996 - 2004	70.076	0.193	-4.501	529384.2	267
Bering Slope	1981 - 1993	66.108	0.149	-8.648	116.423	30
	1996 - 2004	69.269	0.237	-3.479	6052.073	363
Southeast Slope	1981 - 1993	70.818	0.097	-11.369	203.389	30
	1996 - 2004	68.343	0.307	-1.714	13227.5	605

Table 4. Estimated LVB growth curve parameters for female sablefish stratified by region and time period in Alaskan waters.

Area	Time Frame	L_∞	κ	t_0	RSS	n
Chirikof Slope	1981 - 1993	78.151	0.197	-2.659	551.287	26
	1996 - 2004	77.247	0.296	-0.798	13303.94	485
Aleutian Slope	1981 - 1993	74.679	0.185	-3.800	440.07	30
	1996 - 2004	77.804	0.216	-2.267	25557.06	795
Kodiak Slope	1981 - 1993	75.163	0.243	-2.719	896.831	30
	1996 - 2004	78.605	0.314	-0.483	19207.57	602
Shumagin Slope	1981 - 1993	75.379	0.225	-2.552	431.381	28
	1996 - 2004	81.298	0.183	-2.813	1453269	563
Bering Slope	1981 - 1993	69.468	0.241	-3.977	462.229	30
	1996 - 2004	76.380	0.224	-2.692	12149.44	533
Southeast Slope	1981 - 1993	78.854	0.153	-5.348	361.443	31
	1996 - 2004	80.919	0.268	-0.854	19999.35	515

Table 5. Estimated length-weight and weight-at-age relationships for 1996-2004 sablefish specimen data.

	Males	Females
n	4889	5767
	Length-weight	
α	1.24E-05	1.01E-05
β	2.960	3.015
RSS	447.3593054	1044.259777
	Weight-at-age	
W_∞	3.162	5.471
κ	0.356	0.238
t_0	-1.129	-1.387
β	2.960	3.015
RSS	288.874	486.526

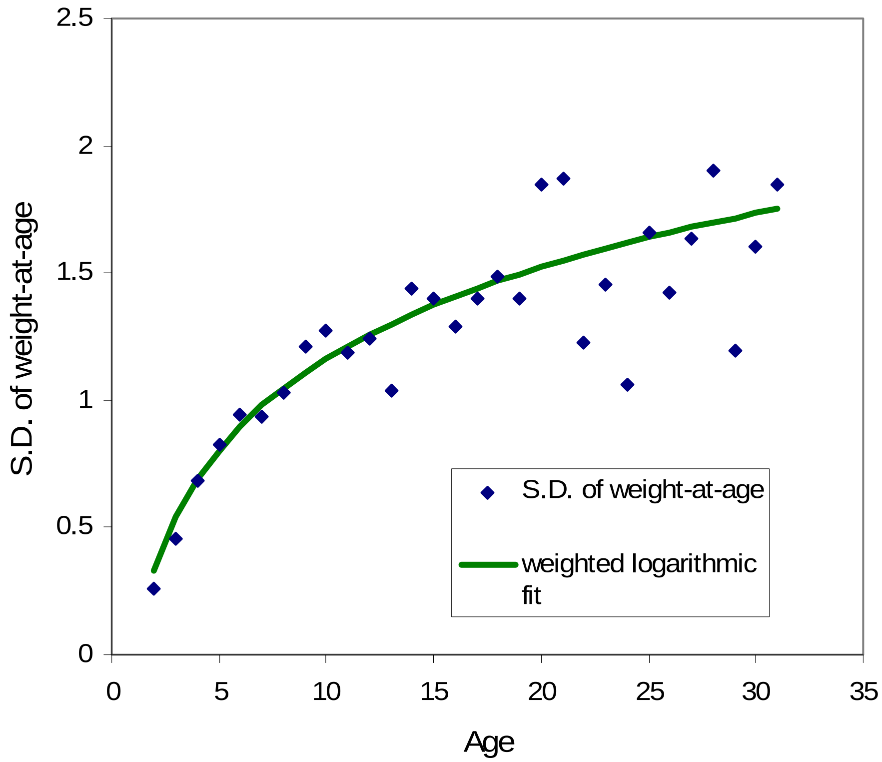

Figure 1. Standard deviation (S.D.) by age of weight for female sablefish.

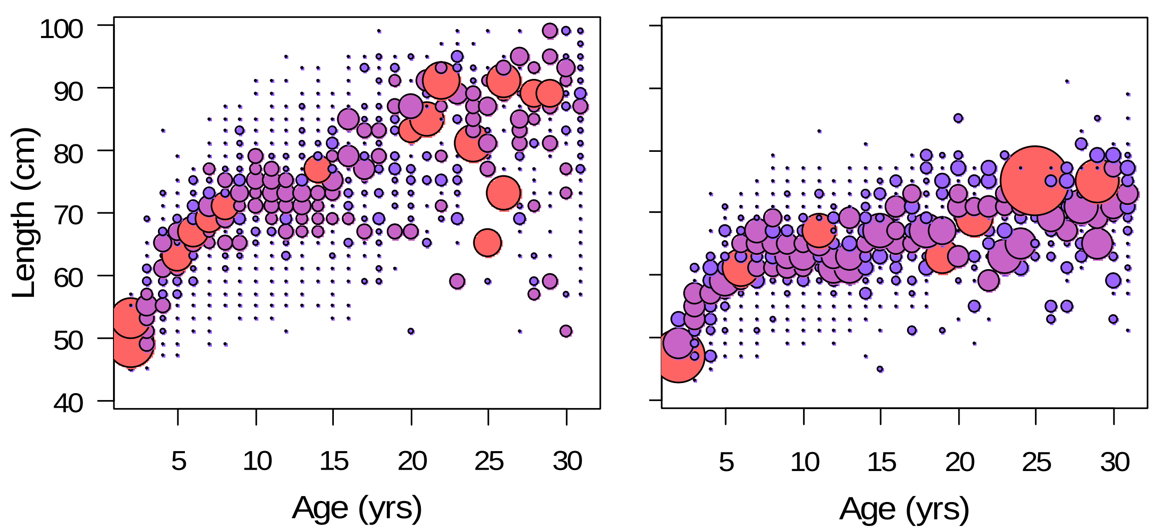

Figure 2. Age length transition matrices using observed lengths-at-age from 1981-1993 for females on left and males on right.

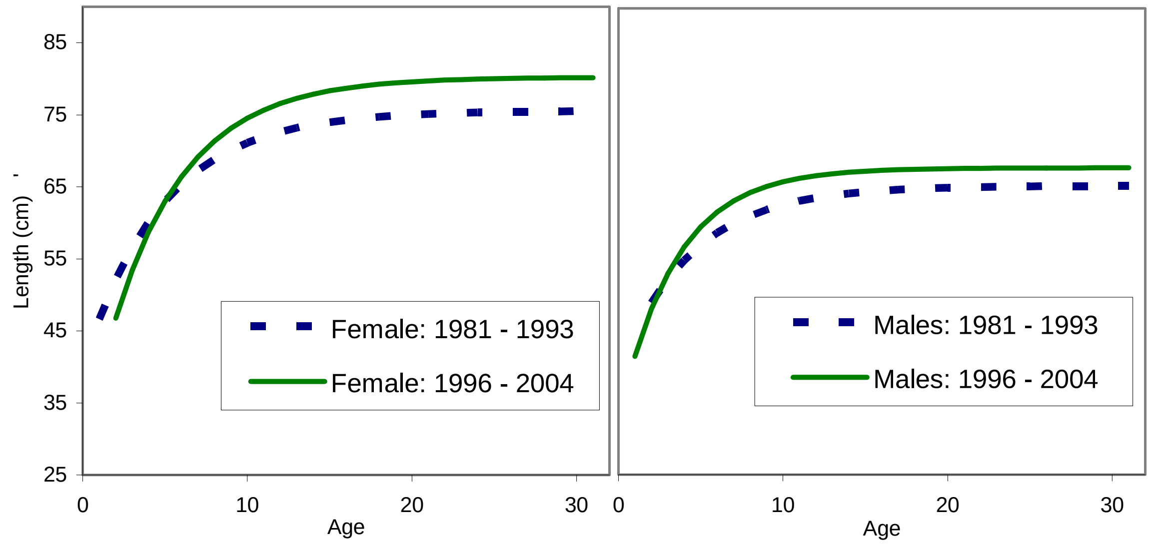

Figure 3: Comparison of sablefish LVB fit to length-at-age from 1981 to 1993 bias-corrected data (blue dashes) and 1996 to 2004 raw data (green solid line) with females shown on the left panel and males on the right.

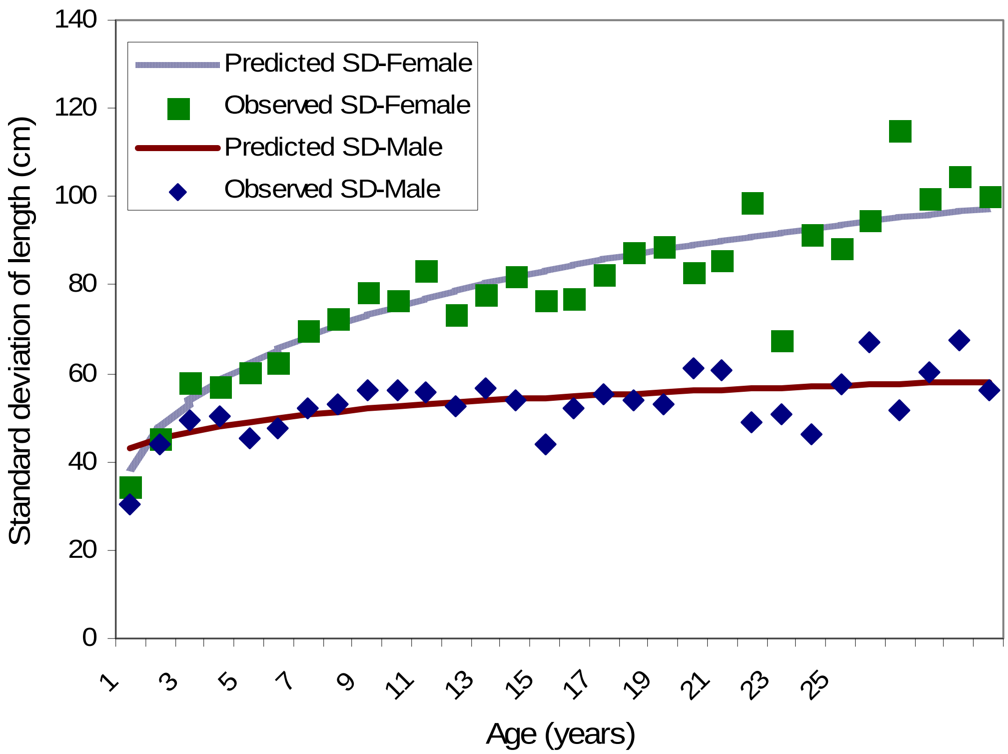

Figure 4. Standard deviations used for normal error in age-length transition matrices.

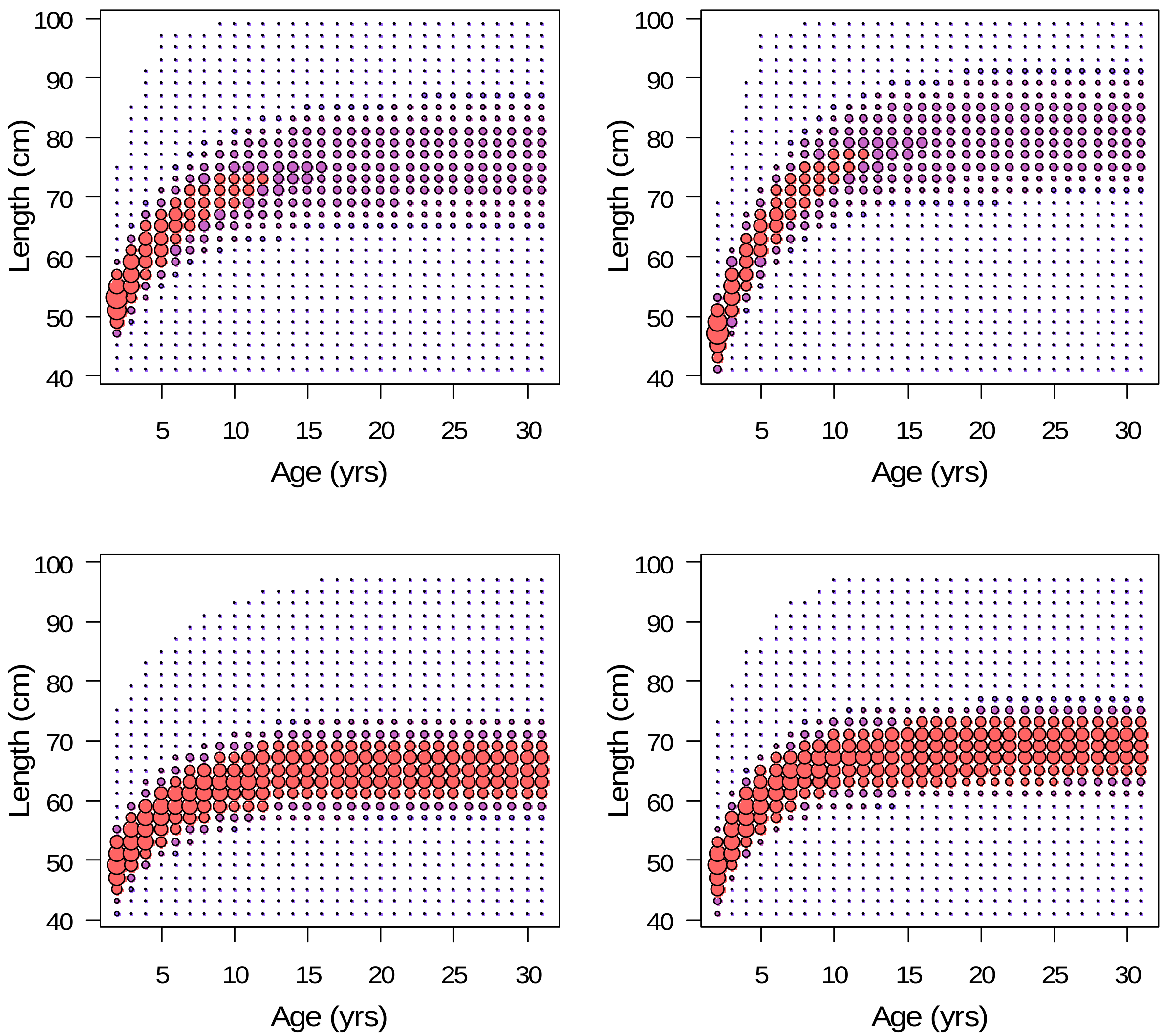

Figure 5. New age-length transition matrices created from new growth analysis for sablefish. Top panels are female, bottom panel are males, left is 1981-1993, right is 1996-2004.

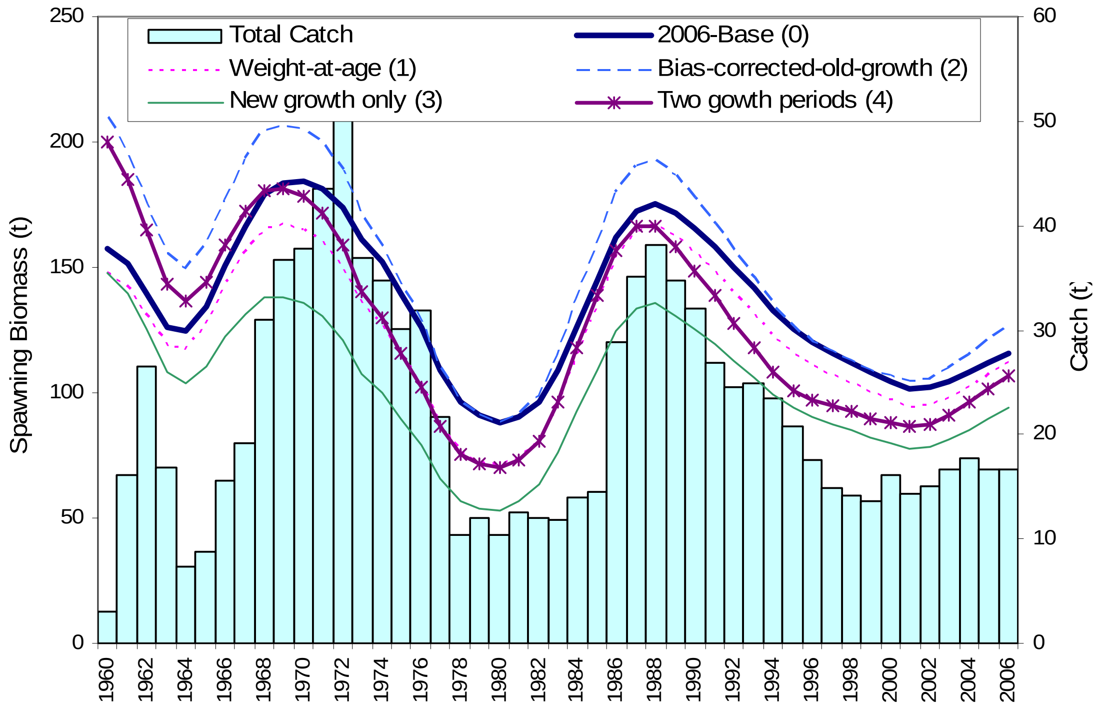

Figure 6. Spawning biomass trajectories for different growth scenarios compared to the 2006 sablefish model. Bars are catch.

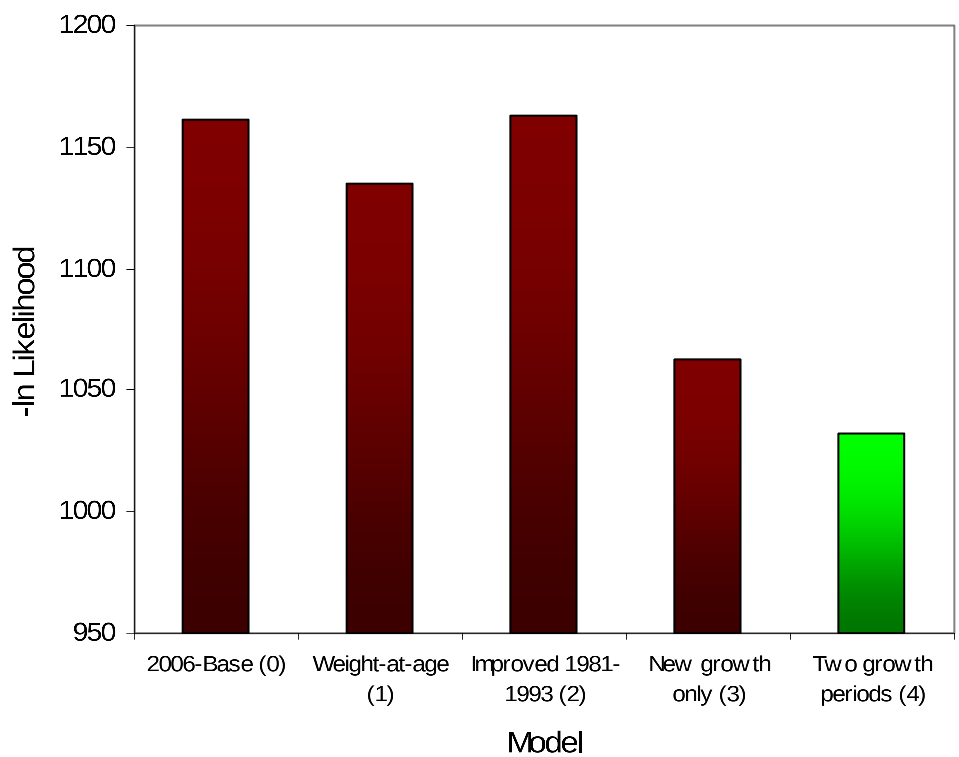

Figure 7. Comparison of fits for different growth scenarios in terms of objective function total (-ln Likelihood).